

Sample collection, Inoculation and Incubation



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Definition

- A specimen may be defined as a small quantity of a substance or object which shows the kind and quality of the whole (sample).
- Specimen collection defined as the collection of the specimen for the purposes of diagnosis, treatment and recovery



Objective of sample collection



- Obtaining **sensitive** and **specific** microbiology results begins with the patient not at the lab.
- Any **error** in specimen collection can lead to **error** in results.

- When the tests are requested, the physician should inform the laboratory staff of the tentative diagnosis (type of infection or infectious agent suspected).

Types of Sample:



- Swabs: Oro-pharyngeal, Ulcer, Burn & Pus

- Body fluids: Whole blood, Serum, Plasma, CSF, Urine, Peritoneal, Pleural & BAL (Bronchoalveolar lavage)



- Tissue: Wound, Skin scraping

- Others: Stool, eye and ear washes

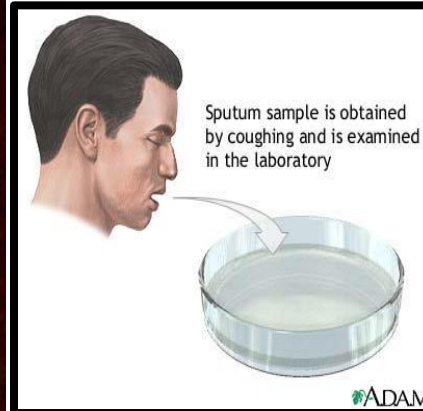


Types of samples

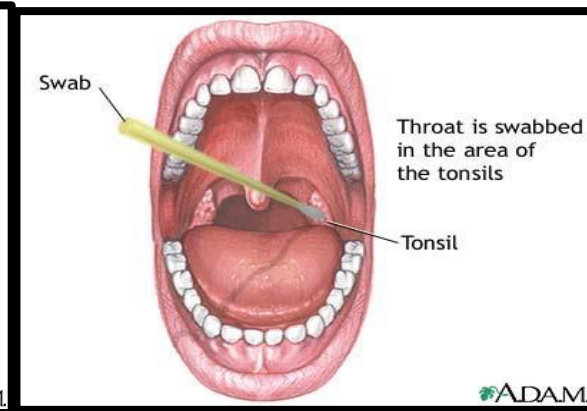
Urine



sputum



swab



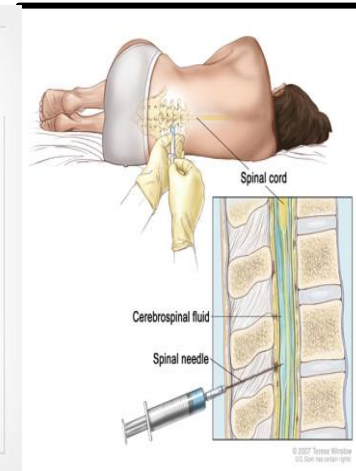
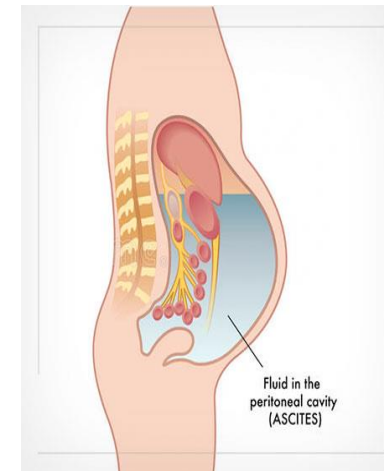
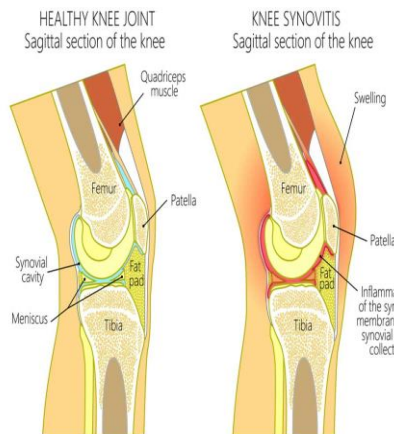
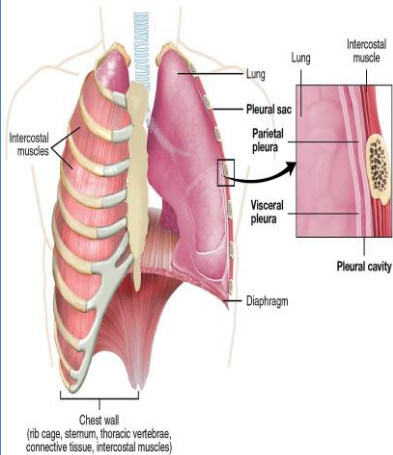
Blood

pleural cavity

joint fluid

peritoneal cavity

CSF



Principles of Sample Collection

- **Consider all safety measures**
- **Avoid / minimize contamination of sample**
- **A specimen must be collected at the optimal time**
- **Patient to be given proper instructions before and during collection**
- **The specimen must be labeled properly**

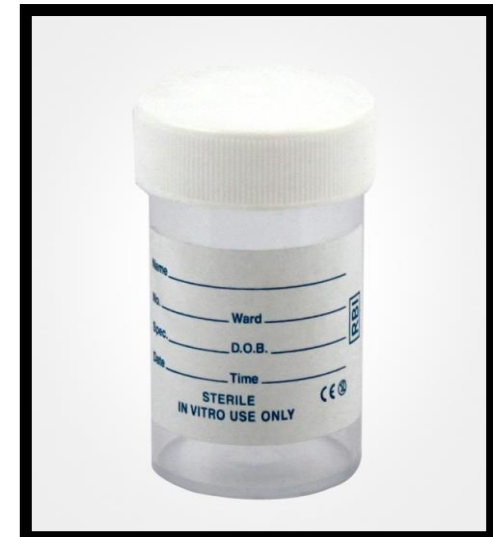


General rules apply to all specimens

- The quantity of materials must be adequate.
- The sample must be representative to the infectious process (sputum, not saliva, pus from underlying lesion, not from its sinus tract; a swab from the depth of the wound, not from its surface)
- contamination of the specimens must be avoided by using only sterile equipment & aseptic technique.
- The specimens must be taken to the laboratory & examined promptly. Special transport media may be needed.
- Before antimicrobial drugs administered; meaningful specimens to diagnose bacterial & fungal infection must be secured.
- If antimicrobial drugs are given before specimens are taken for microbiologic study, drug therapy may have to be stopped & repeat specimens obtained several days later

Labeling Sample

- Date of collection
- Name & Age of patient
- Type of specimen
- Site of collection
- Relevant clinical history
- Whether on antibiotic



- Proper labeling of specimens includes such clinical data as well as the patient`s identifying data (at least two methods of definitive identification) & the requesting physician`s name and pertinent contact information.
- A properly collected specimens is the single most important step in the diagnosis of an infection?
- Selection
- Timing
- Methods of collection of specimens

Transportation of Sample

- Requisition form:
 - Includes necessary details on sample and patient
 - Should be sent to lab with sample asking for the required test
- Preservation of sample:
 - Time
 - Temperature
 - Shipment



Criteria for rejection of specimens

- Unlabeled / mislabeled
- Insufficient volume
- Specimen collected in wrong collection tube
- Contamination
- Inappropriate transport and storage



- Dry shipper Nitrogen

Inoculation of bacteria

- Spreading the sample on the surface of a solid medium
- The purpose is to:
 - Transfer bacteria from the sample to medium
 - Isolate a pure colony of bacteria

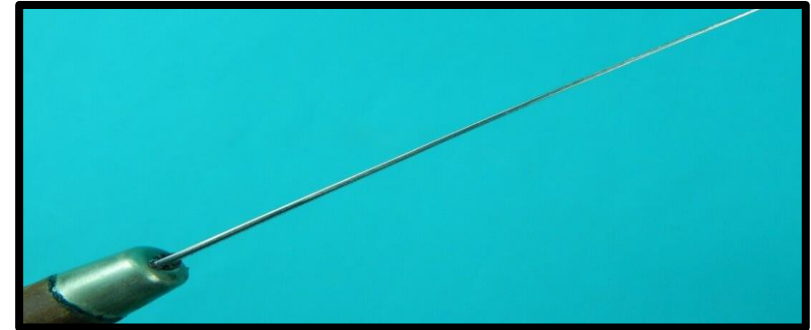


Types of instruments used for inoculation

❖ Inoculation loop



❖ Straight wire



❖ Pasteur pipette

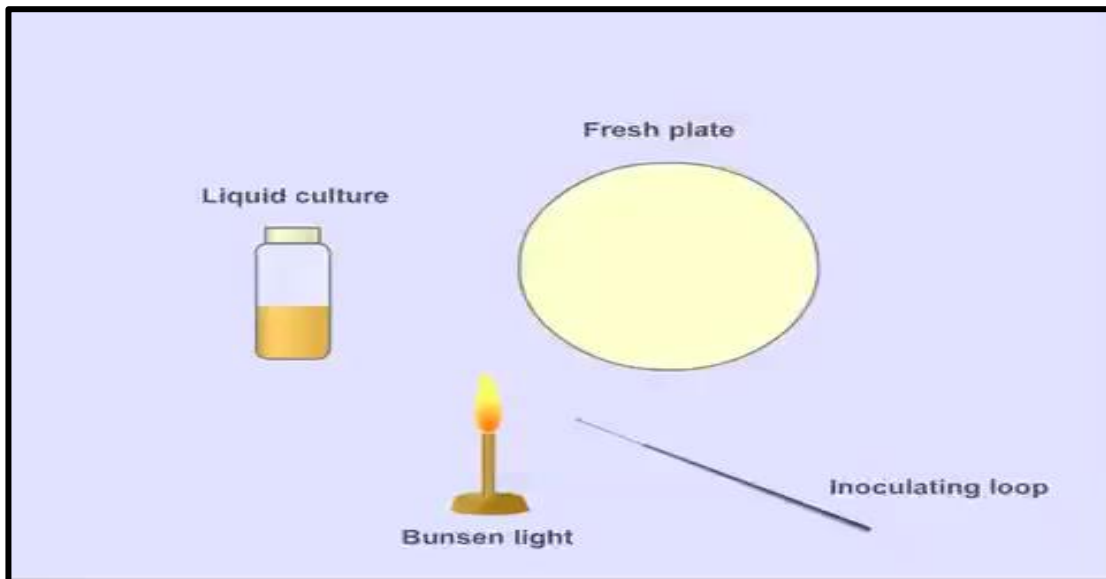


❖ Sterile swab



Types of inoculation

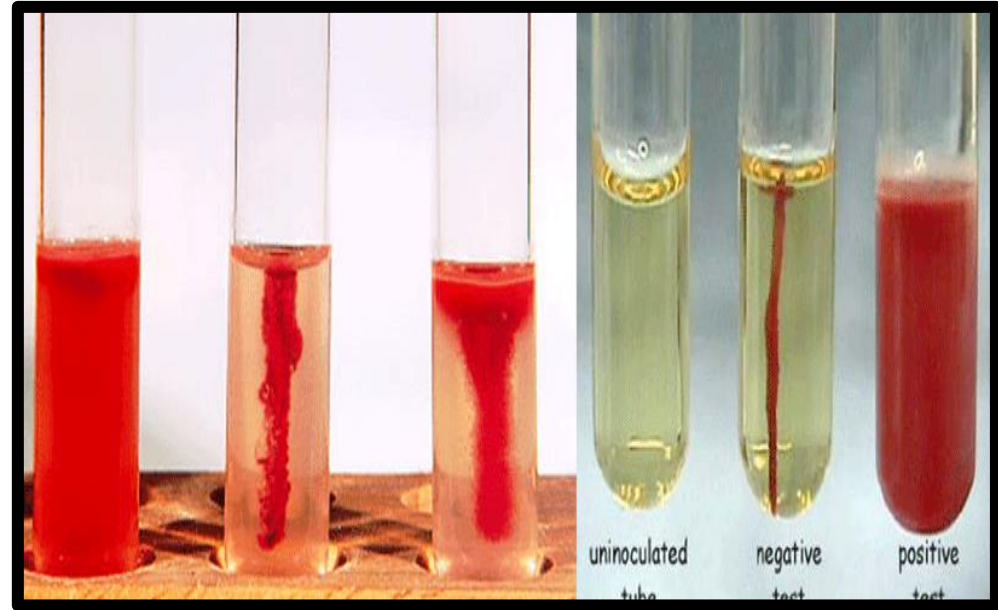
1-Inoculation of fluid media



2-Inoculation of solid media



3-Inoculation of stab media (deep)

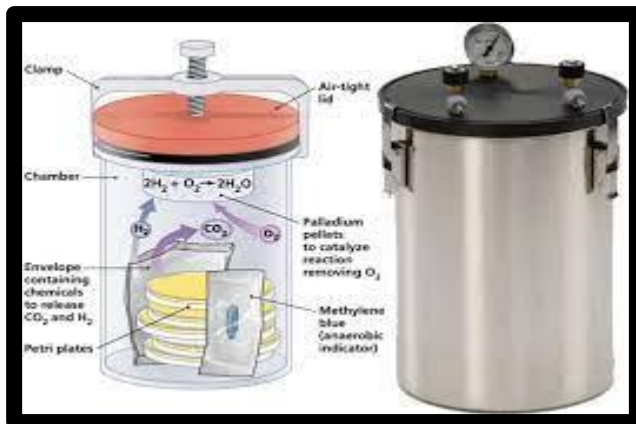


4-Inoculation of slopes (slant) media

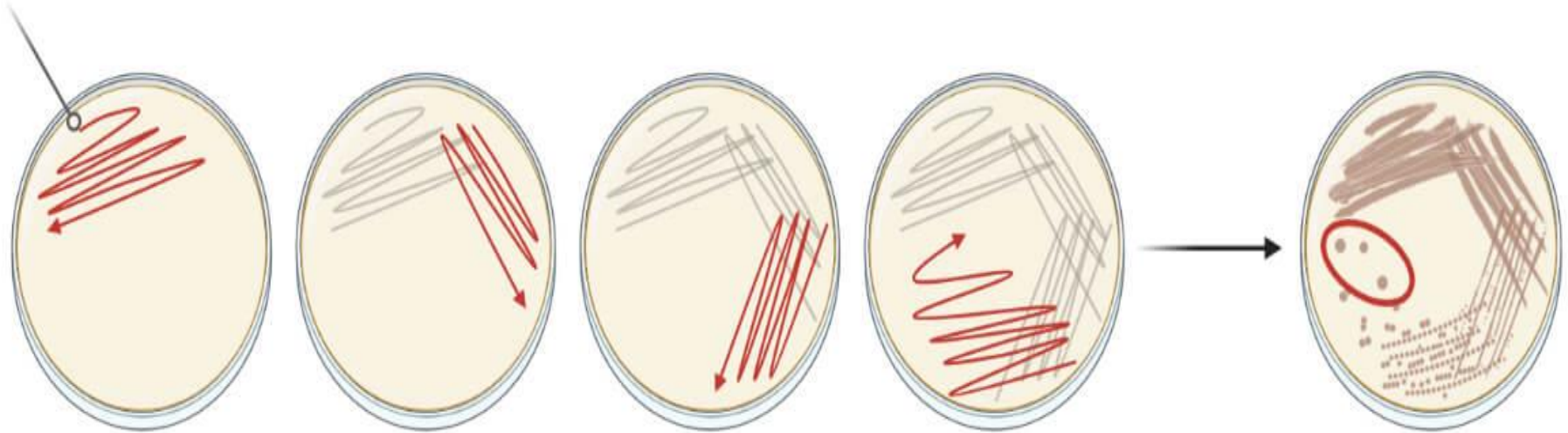


Incubation of cultures

- The cultures are incubated to achieve required temperature and growth environment for a length of time
- Incubator (O_2 and CO_2)
- Anaerobic Jar



Streak Plate Method



Inoculation procedure (flame loop between streaks)

Formation of discrete colonies

- Many pathogenic microorganisms **grow slowly**, & **days** or even **weeks** may elapse before they are **isolated** and **identified**
- Treatment can not be **deferred** until the process is complete.
- The physician should begin treatment with drugs aimed at the organism through to be responsible for the patient`s illness.
- As the laboratory staff begins to **obtain results**, they will inform **health care providers**, who can then **reevaluate** the **diagnosis** & the **therapeutic program**.
- This “feedback” information from the laboratory consists of preliminary reports of the results of individual steps in the isolation & identification of the causative agent.

References

- Luis M. de la Maza, Marie T. Pezzlo, Cassiana E. Bittencourt, Ellena M. Peterson. **2020. Color Atlas of Medical Bacteriology.** (2020, Wiley) - libgen.lc.
- Gary W. Procop, Deirdre L. Church , et al. **2017. Koneman's Color Atlas and Textbook of Diagnostic Microbiology. 7th Edition.** Jones & Bartlett Learning

The image features a dark background with numerous out-of-focus, glowing yellow and orange light spots, creating a bokeh effect. The text "GOOD LUCK" is centered in a white, bold, sans-serif font.

GOOD LUCK